

Vulnerability Scanning with Nmap

One of the most insightful parts of my cybersecurity training was understanding how attackers identify weaknesses long before they strike. Using Nmap, I explored how open ports and running services can reveal critical information about a system's exposure.

In my lab environment, I scanned a Metasploitable machine, a deliberately vulnerable target, to simulate a real-world reconnaissance scenario. By running a simple command like:

```
nmap -sS -sV -p- <target>
```

I discovered open ports such as FTP (21), SSH (22), Telnet (23), and Http (80), etc. Each port reflected potential entry points that, if left unsecured, could lead to exploitation.

This exercise was more than just a technical scan; it was an eye-opener into how proper network auditing and patch management can prevent serious incidents.

The key takeaway is clear, cybersecurity begins with visibility. Knowing what's open, what's running, and what's outdated forms the foundation of defense. Learning to see networks through both defensive and offensive lenses has strengthened my skills and deepened my understanding as an aspiring SOC analyst.